Ascento: A Two-Wheeled Jumping Robot

Victor Klemm*, Alessandro Morra*, Ciro Salzmann*, Florian Tschopp, Karen Bodie, Lionel Gulich, Nicola Küng, Dominik Mannhart, Corentin Pfister, Marcus Vierneisel, Florian Weber, Robin Deuber, and Roland Siegwart

***Abstract*— Applications of mobile ground robots demand high speed and agility while navigating in complex indoor environments. These present an ongoing challenge in mobile robotics. A system with these specifications would be of great use for a wide range of indoor inspection tasks. This paper introduces *Ascento*, a compact wheeled bipedal robot that is able to move quickly on flat terrain, and to overcome obstacles by jumping. The mechanical design and overall architecture of the system is presented, as well as the development of various controllers for different scenarios. A series of experiments[1] with the final prototype system validate these behaviors in realistic scenarios.**

I. Introduction

The emergence of competent mobile robots over the past decade has pushed the development of inspection robotics in research and industry [1], [2]. While flying systems such as drones already show great maneuverability [3], they are very limited in payload and flight time. Ground robots that can navigate quickly and master indoor obstacles are still the topic of ongoing research, and typically lack speed or versatility in their maneuvers.

The locomotion of ground robots can be categorized roughly into two main fields; either a leg- and foot-based [4], [5], [6], [7], [8] or a wheel-based approach [9], [10]. Whilst some walking robots for indoor spaces show great performance overcoming obstacles such as stairs or slippery terrain, they usually still take a significant amount of time to execute these complex movements. In contrast, robots with rotating elements such as wheels are well suited for flat grounds as they can move smoothly, efficiently and fast. However, they usually fail to handle rough terrain, especially if there are obstacles larger than their wheel's radius. An exception are systems with continuous tracks [11]. These can overcome rough terrain and small obstacles with ease but are imprecise and inefficient for turning maneuvers due to slippage.

Robots which combine these two core abilities, fast and smooth maneuvering on flat grounds and dynamic overcoming of obstacles, are rare. Most systems are designed primarily for either one of those tasks, neglecting their performance on the other.

Our goal is to develop an indoor robot that combines the versatility of legs to overcome obstacles by jumping and wheels, which are efficient for fast movement on flat ground.

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[1]Video accompanying paper: `https://youtu.be/U8bIsUPX1ZU`

Fig. 1: Current prototype of the *Ascento* robot.

Tight indoor environments may compromise a robot's mobility to a large extent. Accordingly, the system must come in a highly compact design.

Current systems, which combine jumping and driving, exist and focus either on jumping very high at the cost of repeatability [12], [13] or require a significantly long time to recharge their jumping mechanism [14]. To the best knowledge of the authors, the system presented in [15] is the most similar design to the *Ascento* robot. It manages to combine legs and wheels efficiently for this purpose, but its size renders it unfit for indoor spaces. In this work, we present *Ascento*, a two-wheeled jumping robot in a small form factor especially well-suited for indoor environments. The structural components were created with topology optimization and are fully 3D-printed. An optimized leg geometry decouples the driving and jumping motion and allows the robot to recover from various fall scenarios. Stabilization and driving is achieved through a linear quadratic regulator (LQR) controller. For jumping and fall recovery maneuvers, a sequential feed forward controller with feedback tracking is used. To validate this concept, the real-world prototype demonstrated stabilized driving, jumping and fall recovery in multiple experiments.

The three main contributions can be summarized as:

- The mechanical design of a two-wheeled balancing robot with a parallel elastic jumping mechanism, built from topology optimized parts.
- Dedicated software for controlling the robot.
- Successful experimental evaluation of the idea on a real world prototype.

The remainder of this paper is structured as follows: In section II, we describe the system's mechanical design and list the integrated hardware. In section III, a brief outline

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Ascento：一种双轮跳跃机器人

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摘要——移动地面机器人的应用要求在复杂室内环境中实现高速与敏捷性。这给移动机器人学带来了持续的挑战。具备这些特性的系统将广泛适用于各类室内巡检任务。本文介绍了*Ascento*，一种紧凑型轮式双足机器人，它能够在平坦地形上快速移动，并通过跳跃越过障碍物。文中阐述了该系统的机械设计与整体架构，以及针对不同场景开发的多种控制器。最终原型系统的一系列实验[1] 在真实场景中验证了这些行为。

I. 引言

过去十年间，高性能移动机器人的出现推动了巡检机器人学在研究和工业领域的发展 [1], [2]。尽管无人机等飞行系统已展现出卓越的机动性 [3],，但其有效载荷和飞行时间极为有限。能够快速导航并克服室内障碍物的地面机器人仍是当前研究的课题，且通常缺乏速度或动作的多样性。

地面机器人的运动方式大致可分为两大类：基于腿和脚的方式[4], [5], [6], [7], [8] 或基于轮子的方式 [9], [10]。虽然一些用于室内空间的步行机器人在克服楼梯或湿滑地形等障碍物方面表现出色，但它们通常仍需花费大量时间来执行这些复杂的动作。相比之下，配备轮子等旋转元件的机器人非常适合平坦地面，因为它们能够平稳、高效且快速地移动。然而，它们通常难以应对崎岖地形，尤其是当障碍物大于其轮子半径时。采用连续履带的系统 [11]则是一个例外。这些系统可以轻松克服崎岖地形和小型障碍物，但由于滑移问题，在转弯动作中不够精确且效率低下。

能够结合这两种核心能力——在平坦地面上快速平稳机动以及动态克服障碍物——的机器人十分罕见。大多数系统主要针对其中一项任务进行设计，从而忽略了在另一项任务上的表现。

我们的目标是开发一种室内机器人，它结合了腿的灵活性，能够通过跳跃越过障碍物，并利用轮子在平坦地面上高效快速移动。

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[1]论文配套视频：https://youtu.be/U8bIsUPX1ZU

图1：*Ascento* 机器人的当前原型。

狭窄室内环境可能会在很大程度上影响机器人的机动性。因此，系统必须采用高度紧凑的设计。

当前结合跳跃与驱动的系统存在，但要么专注于以牺牲可重复性为代价实现极高跳跃 [12], [13] ，要么需要极长时间为其跳跃机构充电 [14]。据作者所知， [15] 中提出的系统是与*Ascento* 机器人设计最为相似的方案。它成功地将腿与轮子高效结合以实现此目的，但其尺寸使其不适用于室内空间。在本工作中，我们提出了*Ascento*，一款特别适合室内环境的小尺寸双轮跳跃机器人。结构组件采用拓扑优化设计，并完全通过3D打印制造。优化的腿部几何结构解耦了驱动与跳跃运动，使机器人能够从各种摔倒场景中恢复。稳定与驱动通过线性二次型调节器（LQR）控制器实现。对于跳跃和摔倒恢复操作，则采用带有反馈跟踪的顺序前馈控制器。为验证这一概念，实际原型在多次实验中展示了稳定的驱动、跳跃及摔倒恢复能力。

三个主要贡献可总结如下：

- 采用拓扑优化部件构建的、配备并联弹性跳跃机构的双轮平衡机器人的机械设计。
- 用于控制该机器人的专用软件。
- 在真实世界原型上对该想法进行的成功实验评估。

本文其余部分结构如下：第二节描述系统的机械设计并列出集成硬件。第三节简要概述

of the system's model used for model-based control is given. An explanation of the system's control architecture is given in section IV, highlighting the control strategies used for stabilizing IV-A, jumping IV-B and fall recovery IV-C. The evaluation of real-world experiments is presented in section V. We further list the robot's features which are under development in section VI, before we conclude in section VII.

II. System Description

A. Mechanical Design

The presented system consists of two legs ending with actuated wheels. The legs are attached to a body that houses all electronics as shown in Figure 1. Each leg can be extended and retracted independently by actuating the corresponding motor installed in the hip of the body. This way, the system's total height can be adjusted between $31\,\mathrm{cm}$ and $66\,\mathrm{cm}$. The goal of the leg mechanism is to decouple stabilizing and jumping control as much as possible. This was realized through a three bar linkage which approximates a linear motion of the wheels perpendicular to the ground as seen in Figure 2. By making the line pass through the system's center of mass the leg motion dictates a jump trajectory that does not cause a body rotation. As such a mechanism cannot achieve a perfectly straight line [16], the linkage is optimized numerically for six geometric design parameters with a mean squared error approach to the desired linear trajectory, whereby the optimal lengths and angles of the bars were obtained.

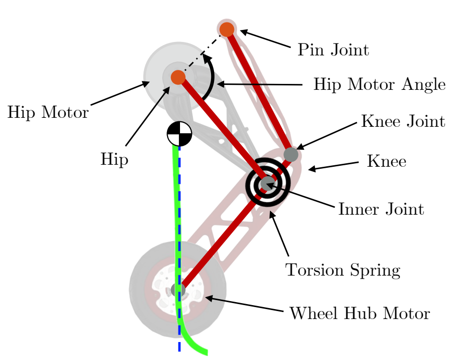

Fig. 2: Left leg assembly and its main components. The optimal linear motion is depicted with a blue dotted line passing through the robot's center of mass and the achieved optimized motion is depicted in green.

To reduce mass and increase the strength of the system the leg components have been designed using topology optimization [17] inspired geometries. Topology optimization is a numerical method that finds a geometry for given set of loads, design space, boundary conditions and a target mass with the goal of minimizing stresses within the structure. The raw topology optimized structure is impractical for post-processing steps and numerical analysis such as finite element method (FEM). The structure was therefore redesigned manually using the optimized structure as template. Manufacturing the optimized parts would be challenging with classical manufacturing processes because of their complex shape. Therefore, all structural parts are 3D-printed from polyamide 12 (PA12) using selective laser sintering (SLS) technology. This technology has also enabled fast prototyping iterations.

To provide smooth stabilization and to counteract disturbances of the tilt angle of the system, near-zero backlash and high wheel torques are required. For this purpose, a custom wheel assembly with frameless hub motors was constructed. This direct drive configuration allows an almost backlash free motion and is very compact. Torsion springs installed in the inner joints as shown in Figure 2 counteract the system's own weight and reduce the control effort of the hip motors when driving or standing, increasing overall efficiency and jumping height.

B. Hardware

The remaining hardware components have been chosen to best fulfill the system's performance requirements. To jump and be able to counteract the spring, when the system has no ground contact, high torques are needed in the hip motor. For this purpose ANYdrive [4] series-elastic actuators are used. They can deliver high peak torques of up to $40\,\mathrm{Nm}$ and have built-in position and torque control. Maxon EC90 frameless electronically commutated (EC) motors with a maximum torque of $3.5\,\mathrm{Nm}$ are utilized in the wheel hub motor. Each wheel motor is equipped with an encoder for precise position and velocity feedback and requires an additional motion controller for torque control. All four motors communicate via a controller area network (CAN) and use an adapter to communicate with the onboard computer. As the main processing unit, which takes full control of the system, an Intel NUC with i7 processor is used. The system is also equipped with an inertial measurement unit (IMU) and a microcontroller to allow communication between the IMU and the computer. Two time-of-flight (ToF) distance sensors used for triggering of a jump are installed next to the wheels. To power all motors, a battery pack composed of four 3-cell lithium-ion polymer (LiPo) batteries connected in series is used. The onboard computer and the remaining electronic devices are powered by a single 4-cell LiPo Battery. A complete component list is shown in Table I.

As a whole, the presented system weights $10.4\,\mathrm{kg}$ and has an operation time of approximately $1.5\,\mathrm{h}$.

C. Software

The control-related software must be computationally efficient to enable high bandwidth controllers, hence all software is written using C++. In addition, the robot operating system (ROS) framework is used for high level communication. A Kalman filter is implemented using sensor data obtained by the IMU and motor encoder measurements. In combination with the model knowledge from III-C, the Kalman filter provides an estimate of the system's state dependent on the hip motor position as described in IV-A. The estimated state information is fed to the controller together with the desired pose from the user as shown in Figure 3. The user input

给出了用于基于模型控制的系统模型。第四节介绍了系统的控制架构，重点阐述了用于稳定（IV-A）、跳跃（IV-B）和跌倒恢复（IV-C）的控制策略。第五节展示了真实世界实验的评估结果。在第七节总结之前，我们还在第六节列出了机器人正在开发中的功能。

II. 系统描述

A. 机械设计

所提出的系统由两条末端带有驱动轮的腿组成。腿连接到一个容纳所有电子元件的身体上，如图1所示。每条腿可以通过驱动安装在身体髋关节处的相应电机独立地伸展和收缩。这样，系统的总高度可以在31厘米到66厘米之间调节。腿部机构的目标是尽可能地将稳定控制和跳跃控制解耦。这是通过一个三杆连杆机构实现的，该机构近似实现了轮子垂直于地面的直线运动，如图2所示。通过使该直线穿过系统的质心，腿部运动决定了跳跃轨迹，从而不会引起身体旋转。由于这种机构无法实现完美的直线运动， [16]，连杆机构通过均方误差方法针对六个几何设计参数进行了数值优化，以逼近理想的直线轨迹，从而获得了最佳的杆件长度和角度。

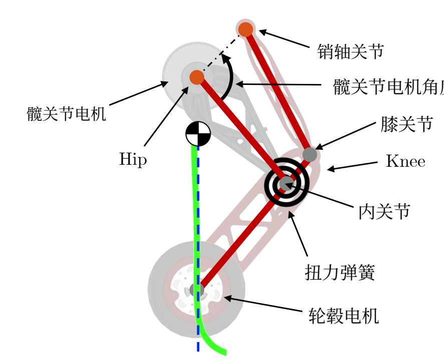

图2：左腿组件及其主要部件。最优直线运动以蓝色虚线表示，该线穿过机器人的质心，而优化后的运动则以绿色表示。

为减轻质量并提高系统强度，腿部组件采用了拓扑优化 [17] 启发式几何结构进行设计。拓扑优化是一种数值方法，它能在给定载荷集、设计空间、边界条件和目标质量的情况下，找到一种几何形状，以最小化结构内部的应力。原始的拓扑优化结构对于后处理步骤和数值分析（如有限元法）来说并不实用。因此，以优化后的结构为模板，对结构进行了手动重新设计。制造这些优化后的部件将因其复杂形状而面临传统制造工艺的挑战。因此，所有结构部件均采用选择性激光烧结技术，使用聚酰胺12进行3D打印。该技术还实现了快速原型迭代。

为了实现平滑的稳定并抵消系统倾斜角的干扰，需要近乎零回差和高车轮扭矩。为此，我们构建了一个采用无框轮毂电机的定制车轮组件。这种直驱配置可实现几乎无回差的运动，且非常紧凑。如图2所示，安装在内关节中的扭力弹簧可抵消系统自身重量，并减少驱动或站立时髋部电机的控制负担，从而提高整体效率和跳跃高度。

B. 硬件

其余硬件组件的选择旨在最佳地满足系统的性能要求。为了在系统无地面接触时实现跳跃并能够对抗弹簧力，髋部电机需要高扭矩。为此，采用了ANYdrive [4] 串联弹性执行器。它们可提供高达40Nm的峰值扭矩，并内置位置和扭矩控制功能。轮毂电机中使用了Maxon EC90无框电子换向电机，其最大扭矩为3.5Nm。每个轮毂电机都配备了一个编码器，用于精确的位置和速度反馈，并需要一个额外的运动控制器来实现扭矩控制。所有四个电机通过控制器局域网进行通信，并使用适配器与机载计算机通信。作为完全控制系统的主处理单元，采用了搭载i7处理器的Intel NUC。系统还配备了惯性测量单元和一个微控制器，以实现惯性测量单元与计算机之间的通信。两个用于触发跳跃的飞行时间距离传感器安装在轮子旁边。为了给所有电机供电，使用了一个由四块3芯锂聚合物电池串联组成的电池组。机载计算机和其他电子设备则由一块单独的4芯锂聚合物电池供电。完整的组件列表见表I。

总体而言，所展示的系统重量为10.4kg，运行时间约为1.5小时。

C. 软件

控制相关软件必须具有高计算效率，以实现高带宽控制器，因此所有软件均使用C++编写。此外，机器人操作系统（ROS）框架用于高层通信。利用惯性测量单元和电机编码器测量值获取的传感器数据，实现了卡尔曼滤波器。结合III-C中的模型知识，卡尔曼滤波器根据IV-A中描述的髋关节电机位置，提供系统状态的估计值。如图3所示，估计的状态信息与用户期望的姿态一同输入控制器。用户输入

TABLE I: Components and suppliers

Component	Name
Wheel Motor	Maxon EC90 Frameless
Wheel Motor Motion Controller	Maxon EPOS4 Compact
Hip Motor	ANYbotics ANYdrive
Wheel Motor USB-to-CAN	ixxat USB-to-CAN V2
Hip Motor USB-to-CAN	Lawicel CANUSB
Motor Battery	Hacker LiPo 5000 mAh 3S
Computer Battery	Turnigy LiPo 2200 mAh 4S
Onboard Computer	Intel NUC KIT NUC7i7BNH
Microcontroller	Arduino Uno
IMU	Analog Devices ADIS16460
Wheel Encoder	AEDL-5810-Z12
ToF Distance Sensors	Terabee TeraRanger Multiflex
3D Mouse	3Dconnexion SpaceMouse
Gesture Control Device	Leap Motion Controller

originates either from a 3D mouse or a gesture control device [18] offering easy and intuitive steering. The controller block includes the stabilizing, jump and fall recovery controllers as well as a high level position controller. Jumping and driving maneuvers are considered decoupled due to the optimized leg geometry and are therefore controlled independently. The stabilizing controller computes and sends torque commands to actuate the wheel motors. Similarly, the jump and fall recovery controllers take full control of the hip and wheel motors.

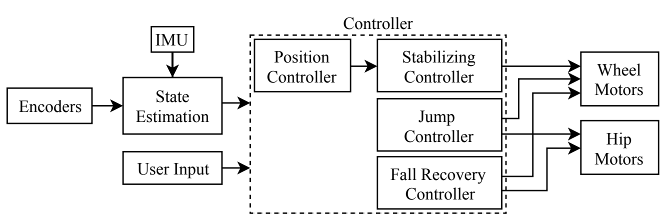

Fig. 3: Overview of the controller architecture.

III. Modeling

To apply model-based control strategies and advanced state estimation techniques a system model to describe the robot's rigid body dynamics is derived.

A. Coordinates and Conventions

1) Notation convention: $\dot{a}$ denotes a temporal derivative, $\hat{a}$ represents an estimate, $\mathbf{A}^\intercal$ a matrix transpose and a_l a symmetrically occurring quantity on the left, a_r on the right, respectively.

2) Generalized coordinates and states: In order to model the simplified system as described in III-C, we introduce a set of system state variables shown in Figure 4. θ denotes the forward tilt angle of the robot, v the planar linear velocity and ω the normal angular velocity. The robot's leg positions are modeled by the sideways tilt angle β (currently only used for an experimental lean mode introduced in section VI) and the distance h to the substitute center of mass. For odometry considerations, we use x, y and γ as planar coordinates for position and orientation and s as the traveled distance on the surface.

We further introduce the generalized coordinates vector $\mathbf{q} = [\theta \quad s \quad \gamma]^\intercal$, which is used in III-C to model the system, and the LQR state vector $\mathbf{x} = [\theta \quad \dot{\theta} \quad v \quad \omega]^\intercal$ for stabilization feedback control as described in IV-A.

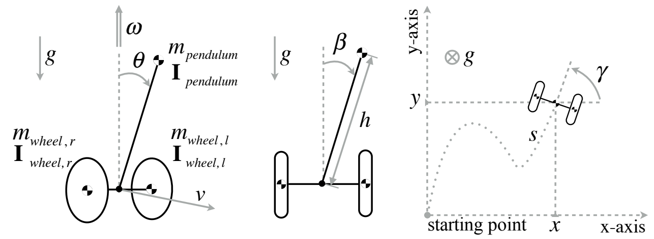

Fig. 4: Generalized coordinates of the simplified *Ascento* robot model, in a diagonal, back and top view. We use m_{body} for mass and $\mathbf{I}_{body}$ for the inertia tensor with the corresponding name of the described body as a subscript. g indicates the Earth's gravitational acceleration.

B. Assumptions

To keep modeling efforts low and the model simple, the following assumptions were made:

1) The dynamics of the leg linkages are neglected.
2) Perfect joints with no friction or hysteresis are assumed.
3) Friction between floor and wheels is simplified by implying a no-slip condition.
4) The motor dynamics are neglected, as they are significantly faster than the rest of the system.
5) System delay is left unmodeled.
6) All links and bodies are rigid.

Assumption 1 implies a fixed hip motor, rendering the model only applicable to a specific leg configuration. This limitation is addressed using interpolated control strategy which lowers the significance of the assumption as described in section IV. All remaining assumptions can successfully be modeled as unknown external disturbances and compensated by a sufficiently fast and robust controller as shown in section V.

C. Rigid Body Model

The assumption of fixed leg geometry reduces the robot's model on a specific height to a two-wheeled inverted pendulum model, consisting of three bodies: Two wheels and an inverted pendulum body with a substitute mass, length and inertia tensor, combined from all included bodies. Using spatial velocity-transport formulae [19], the kinematics of each body are formulated using only the generalized coordinates introduced in III-A. From these expressions, the kinetic and potential energies of all bodies, T and V, respectively, can be derived, leading to the Lagrangian energy function given by

$$\mathcal{L} = T - V. \tag{1}$$

The equations of motion of the system are obtained by using the Lagrange equation of the first kind

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}}\right) - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{J}^\intercal \cdot \mathbf{c} \tag{2}$$

表I：组件与供应商

组件	Name
轮毂电机	Maxon EC90 无框电机
轮毂电机运动控制器	Maxon EPOS4 Compact
髋关节电机	ANYbotics ANYdrive
轮毂电机USB转CAN	ixxat USB转CAN V2
髋关节电机USB转CAN	Lawicel CANUSB
电机电池	Hacker锂聚合物电池5000毫安时3S
计算机电池	Turnigy锂聚合物电池2200毫安时4S
机载计算机	Intel NUC套件NUC7i7BNH
微控制器	Arduino Uno
IMU	Analog Devices ADIS16460
轮式编码器	AEDL-5810-Z12
飞行时间距离传感器	Terabee TeraRanger Multiflex
3D鼠标	3Dconnexion SpaceMouse
手势控制设备	Leap Motion Controller

源自3D鼠标或手势控制设备[18]，提供简单直观的操控。控制器模块包含稳定控制器、跳跃与跌倒恢复控制器以及高层位置控制器。由于优化的腿部几何结构，跳跃和驱动动作被视为解耦，因此独立控制。稳定控制器计算并发送扭矩指令以驱动轮式电机。类似地，跳跃与跌倒恢复控制器完全控制髋关节与轮式电机。

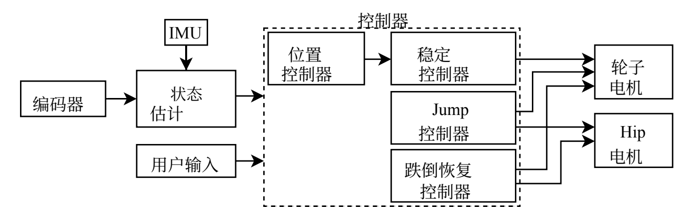

图3：控制器架构概览。

III. 建模

为了应用基于模型的控制策略和先进的状态估计技术，需要推导一个描述机器人刚体动力学的系统模型。

A. 坐标与约定

1) 符号约定：$\dot{a}$ 表示时间导数，$\hat{a}$ 表示估计值，$\mathbf{A}$ 表示矩阵转置，a_l 表示左侧对称出现的量，a_r 表示右侧对称出现的量。

2) 广义坐标与状态：为了对III-C中描述的简化系统进行建模，我们引入了一组如图4所示的系统状态变量。θ 表示机器人的前倾角，v 表示平面线速度，ω 表示法向角速度。机器人的腿部位置通过侧倾角 β（目前仅用于第六节中介绍的实验性倾斜模式）以及到替代质心的距离 h 来建模。出于里程计考虑，我们使用 x、y 和 γ 作为位置和方向的平面坐标，并使用 s 作为表面上的行驶距离。

我们进一步引入广义坐标向量$\mathbf{q} = [\theta \quad s \quad \gamma]^\intercal$（用于III-C中对系统进行建模）以及LQR状态向量$\mathbf{x} = [\theta \quad \dot{\theta} \quad v \quad \omega]^\intercal$（用于IV-A中描述的稳定反馈控制）。

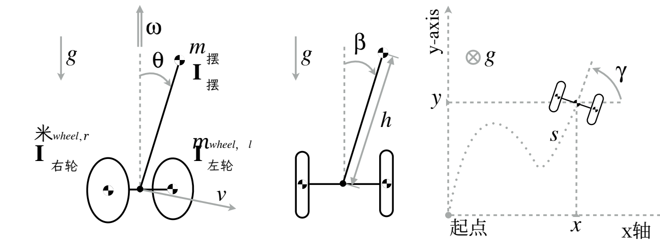

图 4：简化*Ascento*机器人模型的对角、后视和俯视广义坐标。我们使用m_{body} 表示质量，$\mathbf{I}_{body}$ 表示惯性张量，下标为所描述物体的对应名称。 g表示地球重力加速度。

B. 假设

为降低建模难度并保持模型简洁，我们做出以下假设：

1) 忽略腿部连杆的动力学特性。2) 假设为无摩擦、无滞后的理想关节。3) 通过施加无滑移条件来简化地面与轮子之间的摩擦。4) 忽略电机动力学，因其响应速度远快于系统其余部分。5) 系统延迟未纳入建模。
6) 所有连杆和机体均为刚体。

假设1意味着固定髋关节电机，使得该模型仅适用于特定的腿部构型。这一局限性通过采用插值控制策略得以解决，该策略降低了该假设的重要性，如第四节所述。其余所有假设均可成功建模为未知外部扰动，并由足够快速且鲁棒的控制器进行补偿，如第五节所示。

C. 刚体模型

固定腿部几何构型的假设将机器人在特定高度上的模型简化为一个两轮倒立摆模型，该模型由三个刚体组成：两个轮子和一个倒立摆体，该倒立摆体具有由所有包含刚体组合而成的等效质量、长度和惯性张量。利用空间速度传递公式 [19], ，仅使用III-A中引入的广义坐标来表述每个刚体的运动学。根据这些表达式，可以推导出所有刚体的动能 T 和势能 V，从而得到由下式给出的拉格朗日能量函数

$$\mathcal{L} = T - V. \tag{1}$$

系统的运动方程通过使用第一类拉格朗日方程获得

$$\frac{d}{dt}\left(\frac{\partial \mathcal{L}}{\partial \dot{\mathbf{q}}}\right) - \frac{\partial \mathcal{L}}{\partial \mathbf{q}} = \mathbf{J}^\intercal \cdot \mathbf{c} \tag{2}$$

where t is the continuous time variable, $\mathbf{J}$ is the Jacobian transformation matrix of the system and $\mathbf{c}$ is the external Cartesian forces vector. In $\mathbf{c}$, the horizontal and vertical force components of all rigid bodies were set to zero. The torque components were set equal to the particular motor and spring torques acting on the corresponding rigid body.

From Equation 2 the implicit equations of motion

$$\mathbf{M} \cdot \ddot{\mathbf{q}} = \mathbf{f} \tag{3}$$

are derived, with $\mathbf{M}$ being the mass matrix of the system and $\mathbf{f}$ being the forcing term.

All system parameters, such as lengths, masses and moments of inertia, are determined from precise mass measurements of all used components and calculated by using the computer-aided design (CAD) models, assuming constant density of all components.

IV. Control

A two-wheeled robot is inherently unstable. Thus, dedicated control strategies are required not only for jumping, but even standing still and driving. Additionally, to get into operational mode or recover from a fall a specific control maneuver is required.

A. Stabilizing Control

Being able to drive, jump and land again, while staying upright on two wheels, requires a reliable stabilization algorithm. Robustness is also crucial. The robot should be able to handle external disturbances while using as little space as possible to regain its equilibrium. The used approach is an LQR, which is an optimal control strategy for regulating a linear system at minimal cost. Li, Yang and Fan [20] showed that an LQR controller can successfully provide high reliability and robustness for two-wheeled inverted pendulum stabilization applications.

The optimal solution of the LQR's infinite horizon problem is found by solving the discrete-time algebraic Riccati equation

$$\mathbf{F}^\intercal \cdot \mathbf{S} \cdot \mathbf{G} \cdot [\mathbf{R} + \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{G}]^{-1} \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{F} + \mathbf{S} - \mathbf{F}^\intercal \cdot \mathbf{S} \cdot \mathbf{F} - \mathbf{Q} = \mathbf{0} \tag{4}$$

with $\mathbf{F}$ and $\mathbf{G}$ being the discrete-time state-space representation matrices of the system, $\mathbf{Q}$ and $\mathbf{R}$ being the weight matrices and $\mathbf{S}$ being the unknown matrix of the equation.

The optimal feedback gain matrix $\mathbf{K}$ is given by

$$\mathbf{K} = [\mathbf{R} + \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{G}]^{-1} \cdot \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{F}. \tag{5}$$

$\mathbf{Q}$ and $\mathbf{R}$ were selected based on the importance of each state and the desired aggressiveness of the overall system. To simplify this process, the weight matrices were assumed to be diagonal, which reduces the number of adjustable weight parameters to six. To tune these values, intricate tests were performed on the real system.

The choice of the LQR state vector as $\mathbf{x} = [\theta \quad \dot{\theta} \quad v \quad \omega]^\intercal$ omits spatial position and orientation. The idea behind this selection is to leave the problem of position tracking to a dedicated controller which gives direct velocity commands to the LQR. When the operator steers the robot, a reference setpoint is added to v or ω in the state vector before it is fed into the controller, thereby commanding the robot to reach a specific target velocity.

To take into account varying knee angles, the dynamic equations were linearized around ten different, equally spaced leg heights. This yields ten fourth order state-space models and feedback gain matrices $\mathbf{K}$, between which linear interpolation is used. Thereby, the restrictions of the simplified model introduced by assumption 1 in III-B can be resolved.

The system is regulated by the LQR control law

$$\mathbf{u} = -\mathbf{K}(\hat{h}) \cdot \hat{\mathbf{x}} \tag{6}$$

where the input vector $\mathbf{u}$ consists of the left and right wheel torque and $\hat{h}$ is used for linear interpolation between the gain matrices. Here $\hat{\mathbf{x}}$ and $\hat{h}$ are directly supplied by the state observer, as described in II-C. According to the separation principle, this estimation and control setup is guaranteed to be stable and to lead to the robot returning to its perfectly upright equilibrium position as long as noise, disturbances, modeling errors and actuator saturation have no influence on the system's dynamics.

B. Jump Control

The activation of the jump controller triggers a predefined jump sequence which overrides the current drive controllers and takes full control of the robot. The jump controller is a heuristic feed-forward controller, inspired by human jumping motion, with discrete, successive phases (Figure 5). In the following, the jump phases required for the jump on a step are described in further detail.

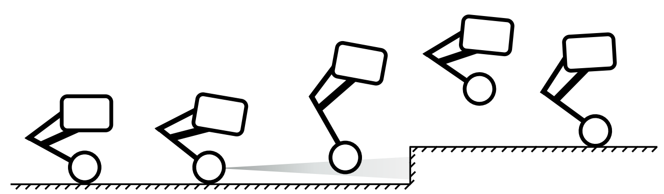

Fig. 5: The discrete phases of the jump sequence. In each phase a different control strategy is applied.

1) Retract Legs: Using a controller that follows a specific trajectory for the hip motors, the robot's legs are retracted. During this process, the stabilizing controller is active.

2) Trigger Jump: As soon as nominal stability after the height change is detected, the robot gathers forward velocity. Using the ToF distance sensors, the following leg extraction is triggered when a predefined distance to the step is reached.

3) Extract Legs: The legs are then extracted by the two hip motors, which are regulated by a proportional integral derivative (PID) controller for synchronous extraction. Once the legs are completely extracted, the stabilizing controller is disabled and ground contact is lost.

其中 t 是连续时间变量，$\mathbf{J}$ 是系统的雅可比变换矩阵，$\mathbf{c}$ 是外部笛卡尔力向量。在 $\mathbf{c}$中，所有刚体的水平和垂直力分量均设为零。扭矩分量被设置为等于作用在相应刚体上的特定电机和弹簧扭矩。

从方程2中，隐式运动方程

$$\mathbf{M} \cdot \ddot{\mathbf{q}} = \mathbf{f} \tag{3}$$

被推导出来，其中$\mathbf{M}$ 是系统的质量矩阵，$\mathbf{f}$ 是力项。

所有系统参数，如长度、质量和转动惯量，均通过对所有使用组件的精确质量测量确定，并利用计算机辅助设计（CAD）模型计算，假设所有组件的密度恒定。

IV. 控制

两轮机器人天生不稳定。因此，不仅需要专门的驱动策略来实现跳跃，甚至需要用于静止站立和驱动。此外，要进入操作模式或从跌倒中恢复，还需要特定的控制动作。

A. 稳定控制

能够在两个轮子上保持直立的同时进行驱动、跳跃和再次着陆，需要一种可靠的稳定算法。鲁棒性也至关重要。机器人应能处理外部干扰，同时尽可能少地占用空间来恢复其平衡。所采用的方法是LQR，这是一种以最小成本调节线性系统的最优控制策略。Li、Yang和Fan [20]表明，LQR控制器能够成功地为两轮倒立摆稳定应用提供高可靠性和鲁棒性。

LQR无限时域问题的最优解通过求解离散时间代数黎卡提方程得到

$$\mathbf{F}^\intercal \cdot \mathbf{S} \cdot \mathbf{G} \cdot [\mathbf{R} + \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{G}]^{-1} \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{F} + \mathbf{S} - \mathbf{F}^\intercal \cdot \mathbf{S} \cdot \mathbf{F} - \mathbf{Q} = \mathbf{0} \tag{4}$$

其中$\mathbf{F}$ 和$\mathbf{G}$ 是系统的离散时间状态空间表示矩阵，$\mathbf{Q}$ 和$\mathbf{R}$ 是权重矩阵，$\mathbf{S}$ 是方程中的未知矩阵。

最优反馈增益矩阵$\mathbf{K}$ 由下式给出

$$\mathbf{K} = [\mathbf{R} + \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{G}]^{-1} \cdot \mathbf{G}^\intercal \cdot \mathbf{S} \cdot \mathbf{F}. \tag{5}$$

$\mathbf{Q}$ 和$\mathbf{R}$ 是根据每个状态的重要性以及整个系统所需的激进性来选择的。为简化此过程，假设权重矩阵为对角矩阵，从而将可调权重参数的数量减少到六个。为了调整这些值，在真实系统上进行了复杂的测试。

选择线性二次型调节器状态向量为$\mathbf{x} = [\theta \quad \theta \quad v \quad \omega]$，省略 ᵀ
了空间位置和方向。其背后的思路是
选择是将位置跟踪问题交给一个专用控制器，该控制器直接发出速度指令

给线性二次型调节器。当操作员操控机器人时，会在状态向量输入控制器之前，向 v 或 ω 添加一个参考设定点，从而命令机器人达到特定的目标速度。

为了考虑变化的膝关节角度，动力学方程围绕十个不同且等间距的腿高度进行了线性化。这产生了十个四阶状态空间模型和反馈增益矩阵$\mathbf{K}$，并在它们之间使用线性插值。由此，可以解决III-B中假设1引入的简化模型的限制。

该系统由LQR控制律调节

$$\mathbf{u} = -\mathbf{K}(\hat{h}) \cdot \hat{\mathbf{x}} \tag{6}$$

其中输入向量$\mathbf{u}$ 由左右轮子的扭矩组成， h 用于增益矩阵之间的线性插值。这里，ˆ$\mathbf{x}$ 和 h 直接由状态观测器提供，如II-C所述。根据分离原理，只要噪声、扰动、建模误差和执行器饱和对系统动力学没有影响，这种估计与控制设置就能保证稳定，并使机器人恢复到完全竖直的平衡位置。

B. 跳跃控制

跳跃控制器的激活会触发一个预定义的跳跃序列，该序列覆盖当前的驱动控制器并完全控制机器人。跳跃控制器是一个启发式前馈控制器，受人类跳跃运动启发，具有离散的连续阶段（图5）。下文将详细描述台阶跳跃所需的跳跃阶段。

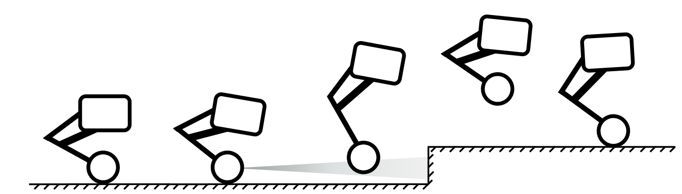

图5：跳跃序列的离散阶段。在每个阶段中应用不同的控制策略。

1) 收腿：使用一个遵循髋关节电机特定轨迹的控制器，机器人的腿被收回。在此过程中，稳定控制器保持激活状态。

2) 触发跳跃：一旦检测到高度变化后的标称稳定性，机器人便积累前进速度。利用飞行时间距离传感器，当达到与台阶的预设距离时，触发随后的腿部伸出。

3) 伸腿：随后，两条腿由两个髋部电机伸出，这两个电机由比例积分微分控制器调节，以实现同步伸出。一旦腿完全伸出，稳定控制器即被禁用，地面接触也随之消失。

4) Fly Phase: A PID loop is used on the hip motor positions with reference on a retracted leg position. Thereby, a virtual spring damper element is simulated. This behaviour is desirable to either jump over high obstacles or prevent the wheels from touching the stair edge.

5) Land: Ground contact is detected when the torques in the hip joints exceed a specific threshold. Upon detection of ground contact, stabilizing control is resumed. Again, a virtual spring-damper element in the hip motors is simulated, allowing for a smooth energy dissipation and controlled landing.

The jump height and forward velocity during the jump can be set by the user via a graphical user interface (GUI). This adjusts the parameters of the jump phases for different scenarios such as jumping on spot, jumping while driving and jumping onto a step.

C. Fall Recovery Control

After a fall or during start-up, the robot is not in its upright position. Stand up procedures are addressed for three out of four resting positions (Figure 6) [21]. Furthermore, the system is able to go into these resting positions in a controlled manner.

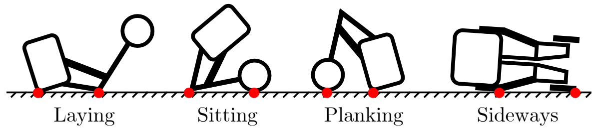

Fig. 6: All four stable positions the robot may fall into. The contact points between robot and ground are represented by red dots.

The resting positions are defined by the contact points between ground and robot. In the laying position, the robot contacts the ground with the legs and rear parts of the body. When in sitting position, it touches the ground with the legs and wheels and in the planking position with the front part of the body and wheels. When the robot lays on its side contacting a single leg and wheel (sideways position), it is neither able to recover from nor achieve this resting position in a controlled manner.

Similar to the jump procedure, the stand up procedure is composed of discrete, successive phases (Figure 7) which are similar for all resting positions. The event is triggered by the user and overrides the current drive controller.

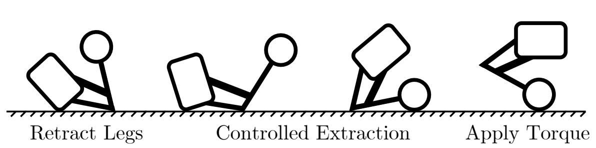

Fig. 7: Phases of the stand up procedure. The controlled extraction step only applies for the laying position.

1) Retract Legs: Using a controller that follows a specific trajectory for the hip motors, the robot's legs are retracted.

2) Controlled Extraction: This step only applies for the laying position. Following a predetermined trajectory, the legs are extracted and later retracted again. This induces a rotation around the knee, and the robot achieves the sitting position. A controlled extraction of the legs is required as too fast an extraction would make the robot lose ground contact and have a hard impact. On the other hand, too slow an extraction would not suffice to tip the robot over to the sitting position.

3) Apply Torque: A constant torque is applied to the wheel motors, backwards for the sitting position and forwards for the planking position. The constant torque is applied until the robot has enough rotational energy to reach zero tilt angle. Once the robot stands vertically, the stabilizing controller is turned on and the robot brakes to reach its upright idle position.

Entering a resting position in a controlled manner is achieved by turning off the stabilizing controller and applying a small torque to the wheel motors to control the fall direction.

V. Experiments

A. Simulation

All the control algorithms were tested in a simulation, based on Gazebo [22] as a physics engine. A model of the robot with mass and inertia values from the CAD model allows for a realistic and computationally efficient testing environment. The similar behaviour of the simulation and the experiments suggests the validity of subsection III-B and allows for further development section VI.

B. Experimental Results

In this section, results from a series of experiments with the prototype system are presented. The dimensions and other technical specifications of the robot are shown in Figure 8 and Table II, respectively. Multiple experiments are presented to demonstrate the robot's specific capabilities regarding stabilizing performance (V-B.1), jumping (V-B.2) and fall recovery (V-B.3).

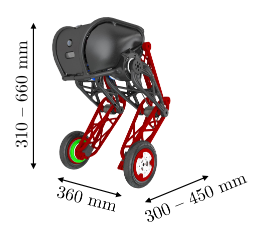

Fig. 8: Main system dimensions.

TABLE II: Technical specifications of the robot.

Category	Value
Weight	10.4 kg
Max. linear velocity	$8.0\,\mathrm{km\,h^{-1}}$
Max. angular velocity	$1.1\,\mathrm{rad\,s^{-1}}$
Max. jumping height	0.4 m
Battery lifetime	1.5 h

1) Stabilizing Performance: The goal of the system is to stay upright at all times. Without compromising stability, the system is able to sustain large external disturbances. The robot's response to a hit from a wooden stick is shown in Figure 9. Beside impulsive disturbances, the system is also able to go back to its equilibrium position if permanent or longer-lasting disturbances are applied to the system (pulling, pushing or adding weight).

4) 飞行阶段：在髋关节电机位置上使用PID控制回路，并以缩回的腿部位置作为参考。由此模拟出一个虚拟的弹簧阻尼元件。这种行为有助于跳过较高的障碍物，或防止轮子碰到楼梯边缘。

5) 着陆：当髋关节的扭矩超过特定阈值时，检测到地面接触。检测到地面接触后，恢复稳定控制。同样，在髋部电机中模拟了一个虚拟的弹簧阻尼元件，从而实现平滑的能量耗散和受控着陆。

跳跃高度和跳跃过程中的前进速度可由用户通过图形用户界面（GUI）设置。这会调整跳跃阶段的参数，以适应不同场景，例如原地跳跃、行进中跳跃以及跳上台阶。

C. 跌倒恢复控制

跌倒后或启动时，机器人并不处于直立位置。针对四种静止位置中的三种（图6） [21]，设计了站立程序。此外，系统能够以受控方式进入这些静止位置。

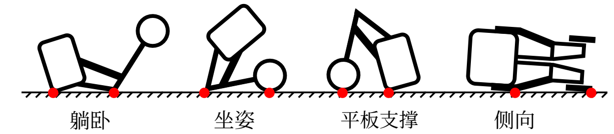

图6：机器人可能跌入的四种稳定位置。机器人与地面之间的接触点用红点表示。

静止位置由机器人与地面之间的接触点定义。在躺卧位置，机器人通过腿和身体后部接触地面。在坐姿位置，它通过腿和轮子接触地面；在平板支撑位置，则通过身体前部和轮子接触地面。当机器人侧卧，仅接触一条腿和一个轮子（侧卧位置）时，它既无法从该静止位置恢复，也无法以受控方式达到该位置。

与跳跃程序类似，站立程序由离散的连续阶段组成（图7），这些阶段对所有静止位置都相似。该事件由用户触发，并覆盖当前的驱动控制器。

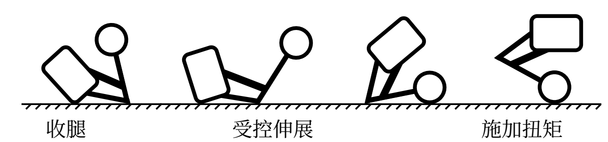

图7：站立程序的各个阶段。受控伸展步骤仅适用于躺卧位置。

1) 收腿：通过使用遵循髋部电机特定轨迹的控制器，机器人的腿被收回。

2) 受控伸展：此步骤仅适用于躺卧位置。按照预定轨迹，腿先伸展随后再次收回。这会引起围绕膝关节的旋转，使机器人达到坐姿位置。需要进行受控的腿部伸展，因为

过快的抽出会导致机器人失去地面接触并产生剧烈冲击。另一方面，过慢的抽出则不足以使机器人翻转到坐姿位置。

3) 施加扭矩：对轮式电机施加恒定扭矩，坐姿位置时向后，平板支撑位置时向前。持续施加恒定扭矩，直到机器人拥有足够的旋转能量达到零倾斜角。一旦机器人垂直站立，稳定控制器启动，机器人制动以回到直立空闲位置。

通过关闭稳定控制器并对轮式电机施加一个小扭矩来控制倾倒方向，从而以受控方式进入休息位置。

V. 实验

A. 仿真

所有控制算法均在基于Gazebo [22] 作为物理引擎的仿真环境中进行了测试。使用来自CAD模型的具有质量和惯性值的机器人模型，可以构建一个逼真且计算高效的测试环境。仿真与实验的相似行为验证了第III-B节的有效性，并为第六节的进一步开发奠定了基础。

B. 实验结果

本节展示了使用原型系统进行的一系列实验结果。机器人的尺寸及其他技术规格分别如图8和表II所示。通过多项实验来展示机器人在稳定性能（V-B.1）、跳跃（V-B.2）和跌倒恢复（V-B.3）方面的特定能力。

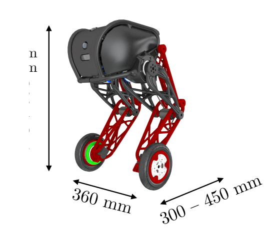

图8：系统主要尺寸。

表II：机器人的技术规格。

类别	数值
重量	10.4 公斤
最大线速度	$8.0\,\mathrm{km/h^{-1}}$
最大角速度	$1.1\,\mathrm{rad/s^{-1}}$
最大跳跃高度	0.4 米
电池寿命	1.5 小时

1) 稳定性能：系统的目标是始终保持直立。在不影响稳定性的前提下，系统能够承受较大的外部干扰。机器人对木棍击打的响应如图9所示。除了脉冲式扰动外，如果对系统施加永久性或持续时间更长的扰动（如拉、推或增加重量），系统也能够恢复到其平衡位置。

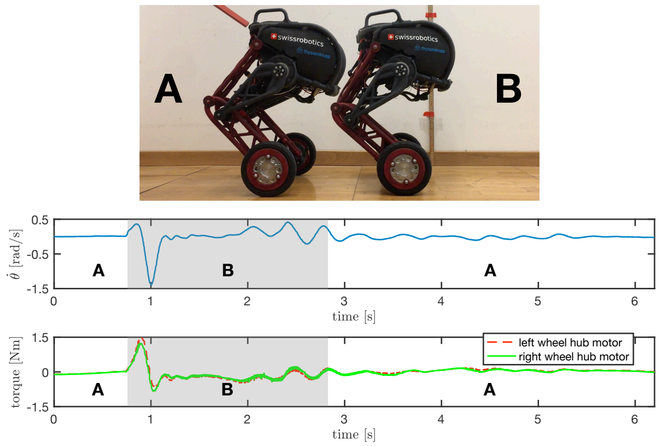

Fig. 9: Image: The system on the left (A) is in an unstable equilibrium position and about to be disturbed by a red stick. On the right side (B) the robot is recovering from the disturbance. Graphs: The top graph shows the angular velocity $\dot{\theta}$ response, and the bottom one shows the left and right wheel motor torque responses $\mathbf{u}$ over time.

2) Jumping: The system is able to jump on small steps with a height of 10 cm as demonstrated in Figure 10. The robot needs at least 90 cm both before and after a step to accelerate to the target velocity and in order to land safely.

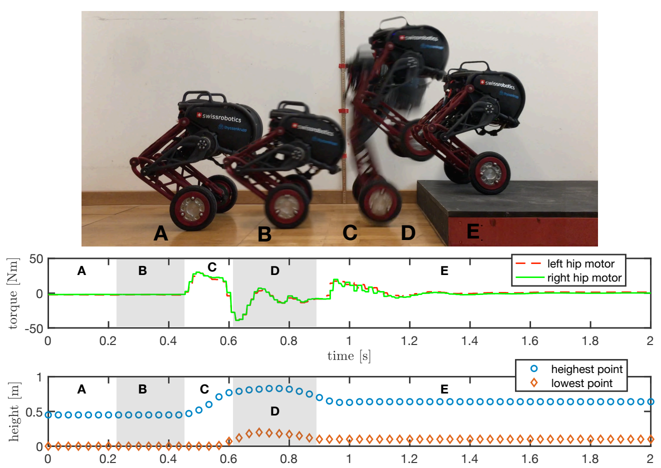

Fig. 10: Image: The jump of the system onto a real step with the five phases: Retract Legs (A), Trigger Jump (B), Extract Legs (C), Fly Phase (D) and Land (D). Graphs: In the top graph the hip torques and in the bottom the highest and lowest points of the system are plotted over time during a jump cycle.

3) Fall Recovery: The standing up procedure from all recoverable resting positions requires less than 2 m of space. In Figure 11 the stand up procedure is shown for the laying position.

VI. Experimental Features

Besides the capabilities tested on the real-world prototype, the system also has some experimental features which are to

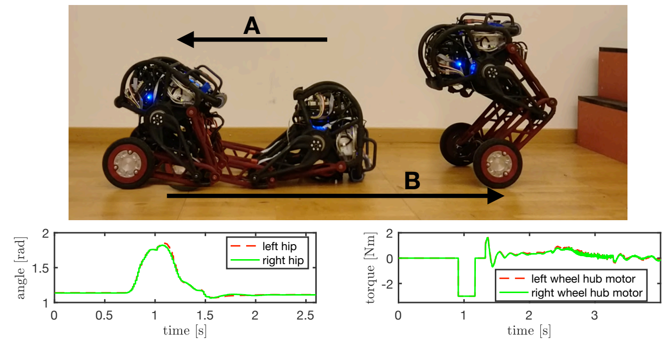

Fig. 11: Image: The system is able to stand up by first getting itself into a sitting position (A) and then to the standing position (B). Graphs: The left graph shows the hip motor angle movements in order that the system sits up (A). The right graph represents the wheel motor torque commands for standing up (B).

this date only validated in the simulation environment. An exploration algorithm was developed which can, depending on a local and global planning strategy, map a previously unknown environment. Using the occupancy information of such a map, the robot is able to find and follow a collision-free path from its current position to any desired destination [23], [24]. Additionally, a force-compensating lean mode is introduced to allow the system to be able to turn faster without tipping sideways. This is done by varying each leg angle individually to tilt the robot inwards in β. The system's unique design and modular setup make it also an interesting object for further academic investigation, e.g., developments of new control strategies such as balancing on one leg [25] or model predictive control (MPC).

VII. Conclusion

This work presented the *Ascento* platform, a two-wheeled balancing robot that is able to navigate quickly on flat surfaces and to overcome obstacles by jumping. The system's topology optimized, 3D-printed mechanical design has proven to be both lightweight and impact-resistant. Additionally, a robust, model-based LQR controller has been successfully implemented on the prototype system. In multiple experiments with the prototype system we demonstrated autonomous jumping onto a step and the ability to recover from downfalls into various positions. Finally, the robot's operating environment could be extended from indoor to outdoor by incorporating terrain-adaptive principles of legged robotics [26].

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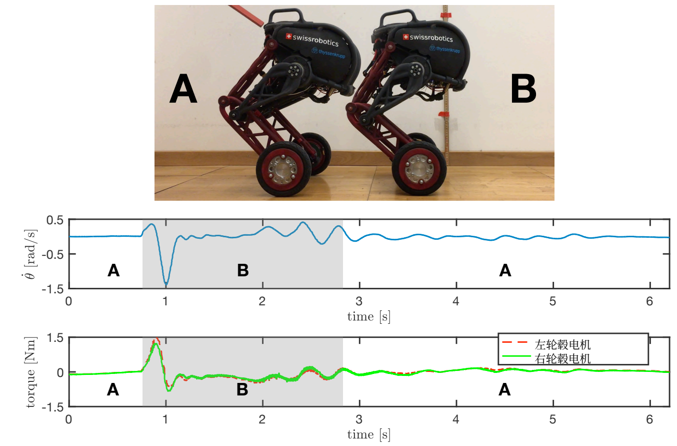

图9：图像：左侧（A）的系统处于不稳定平衡位置，即将被一根红色棍子扰动。右侧（B）的机器人正在从扰动中恢复。图表：顶部图表显示了角速度 $\dot{\theta}$ 响应，底部图表显示了左右轮毂电机扭矩响应随时间的变化$\mathbf{u}$ 。

2) 跳跃：系统能够跳跃到高度为10厘米的小台阶上，如图10所示。机器人需要在台阶前后至少保持90厘米的距离，以便加速到目标速度并安全着陆。

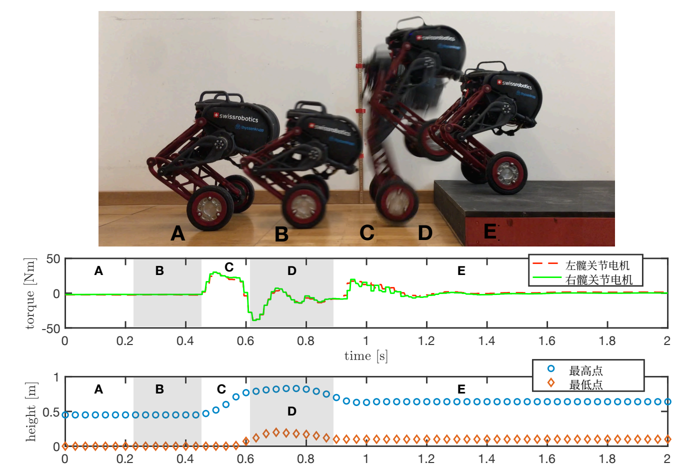

图10：图像：系统跳跃到一个真实台阶上的五个阶段：收腿（A）、触发跳跃（B）、伸腿（C）、飞行阶段（D）和着陆（D）。图表：顶部图表显示了跳跃周期中髋关节扭矩随时间的变化，底部图表显示了系统的最高点和最低点。

3) 跌倒恢复：从所有可恢复的静止位置执行站立过程所需空间小于2米。图11展示了从躺卧位置进行站立的过程。

VI. 实验特性

除了在真实世界原型上测试过的功能外，该系统还具备一些实验特性，这些特性截至

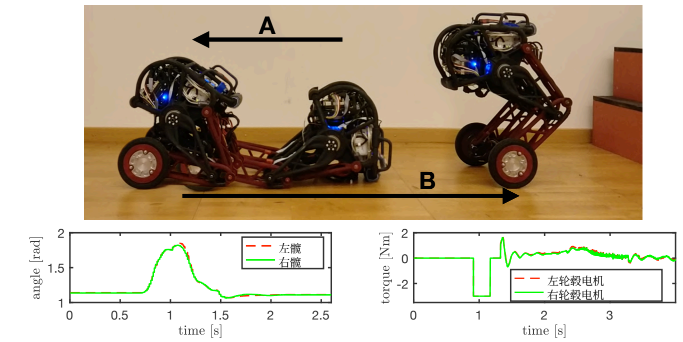

图11：图像：系统能够通过先进入坐姿位置（A），然后进入站立位置（B）来实现站立。图表：左图显示了系统坐起（A）过程中髋关节电机角度的运动。右图展示了站立（B）过程中轮毂电机扭矩指令。

目前仅在仿真环境中得到验证。我们开发了一种探索算法，该算法能够根据局部与全局规划策略，对先前未知的环境进行地图构建。利用此类地图的占据信息，机器人能够找到并遵循一条从当前位置到任意目标点的无碰撞路径[23], [24]。此外，还引入了一种力补偿倾斜模式，使系统能够在不侧翻的情况下更快地转弯。这是通过单独调整每个腿部角度，使机器人在 β中向内倾斜来实现的。该系统的独特设计和模块化结构也使其成为进一步学术研究的有趣对象，例如开发新的控制策略，如单腿平衡 [25]或模型预测控制（MPC）。

VII. 结论

本文介绍了Ascento平台，这是一种双轮平衡机器人，能够在平坦地面上快速移动，并通过跳跃越过障碍物。该系统的拓扑优化、3D打印机械设计已被证明兼具轻量化和抗冲击性。此外，一个鲁棒的基于模型的LQR控制器已成功在原型系统上实现。通过原型系统的多次实验，我们展示了自主跳跃到台阶上的能力，以及从各种跌倒位置恢复的能力。最后，通过融合腿式机器人的地形自适应原理，该机器人的运行环境已从室内扩展到室外 [26]。

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